Microeconomics In-class Quiz 3
Fall 2026

Student ID: _____ **Name:** _____

Instructions

1. Do NOT flip over this page until every student receives this quiz. Your TA will let you know when you can start.
2. During this closed-book quiz, you cannot consult any materials.
3. It is okay to write your reasoning in Korean.
4. There are five questions in total. Both sides are printed, so make sure you check the backside of each page.
5. [**IMPORTANT**] Make your answers **legible**. Clearly delineate your scratchpad work from your final answers. Deducted points due to illegible writing cannot be the reason for reevaluation.

Honor Code: Cheating on exams or quizzes, plagiarizing someone else's answers as one's own, or any other instance of academic dishonesty violates the standards of academic integrity.
Confidentiality Code: Sharing the information of the exam or quiz contents with other students in any form and medium is strongly prohibited, as it raises information inequity.

I, _____, consent to the Honor Code and the Confidentiality Code.
(write your name)

1. A major theme park is analyzing ticket sales for its upcoming summer festival. The manager estimates the daily demand for Locals to be $q^L = 900 - 10p$, and the demand for Tourists to be $q^T = 1100 - 10p$. The manager is considering charging different entry fees for the two groups (third-degree price discrimination). The marginal cost of an additional visitor is \$20.

(a) Derive the marginal revenue (MR) functions for Locals and Tourists.

(b) Find the profit-maximizing price and ticket sales for each group.

(c) Suppose local regulations prohibit price discrimination, forcing the theme park to charge a single uniform price. The aggregate demand is given by:

$$Q = \begin{cases} 1100 - 10p & \text{if } p > 90 \\ 2000 - 20p & \text{if } p \leq 90 \end{cases}$$

Find the profit-maximizing single price and the total number of ticket sales.

2. Alice and Bob play a simultaneous-move game described by the following payoff matrix:

		Bob		
		L	C	R
Alice	T	6, 6	8, 4	4, 3
	M	7, 2	3, 5	2, 9
	B	9, 8	10, 1	5, 2

(a) What is Bob's payoff when Alice plays T and Bob plays C?

(b) Are there strictly dominant strategies in this game? If so, describe them.

(c) Is this game dominance solvable? If so, describe the final action profile. If not, explain why.

3. Are there pure-strategy Nash equilibria in each of the following games? If so, describe all of them.

(a)		Firm B	
		Enter	Stay Out
Firm A	Enter	-5, -5	10, 0
	Stay Out	0, 10	0, 0

(b)		Defender	
		Block Left	Block Right
Striker	Kick Left	0, 10	10, 0
	Kick Right	10, 0	0, 10

(c)		Colleague	
		Slack	Work
You	Slack	2, 2	6, 1
	Work	1, 6	5, 5

(d)		TechCo		
		Aggressive	Moderate	Passive
DataInc	Aggressive	3, 3	6, 2	5, 4
	Moderate	5, 8	5, 6	3, 7
	Passive	4, 6	9, 7	4, 4

4. StreamCo offers two digital subscription services: a Movie Pass (M) and a Sports Pass (S). Assume the marginal cost of providing these services is zero. Two customers, Alex and Jordan, want to purchase these services (either 1 unit or nothing). Alex is willing to pay \$10 for M and \$8 for S. Jordan is willing to pay \$4 for M and \$6 for S. When the price is equal to or less than their willingness to pay, they buy the service.

(a) If StreamCo sets the price for M at \$10 and S at \$8, what total profit is generated?

(b) What prices maximize profit when selling the two passes separately?

(c) If bundling M and S together as a single package, what bundle price should be set to maximize profit, and what is the new total profit?

5. The market for high-end gaming motherboards is dominated by two firms: Alpha and Beta. The inverse market demand is given by $P = 1200 - 3Q$, where Q is the sum of Alpha's output (q_A) and Beta's output (q_B). The marginal cost of producing a motherboard is \$300 for both firms.

(a) Given Beta's production level q_B, derive Alpha's best response function (reaction curve).

(b) Describe the Cournot-Nash equilibrium when the two firms decide their output levels simultaneously.

(c) Consider a Stackelberg sequential-move game where Alpha decides its output level first, and then Beta chooses its output. What is the Alpha's quantity in the Subgame-Perfect Equilibrium?

(d) Suppose Alpha becomes a monopolist because Beta exits the market. What would be the output level in this monopoly?